Life Sciences and
Facility Management

Institute of
Computational Life Sciences

Introduction to Digital Image Processing

Part 1: Objectives and Context

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Objectives

Objectives of digital image processing

- Build machines that can see
- This involves many aspects:
  - Acquire and represent image data
  - Transform and filter images
  - Enhance image data, or restore missing data
  - Optimize information content (compression)
  - Extract information from image data (sensing)
  - Understand image content (semantics)
  - Decide based on image data
  - Synthesize images

Image acquisition

Image transformation

Image enhancement / restoration

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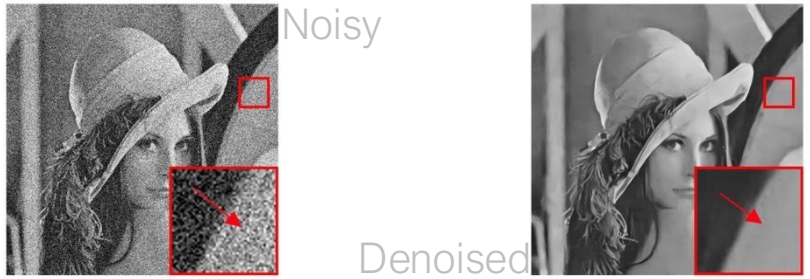
Image filtering



Extract information / features

Original | Edge and... | corner detection

Segmentation

Original | Detection | Segmentation

Objectives of digital image processing

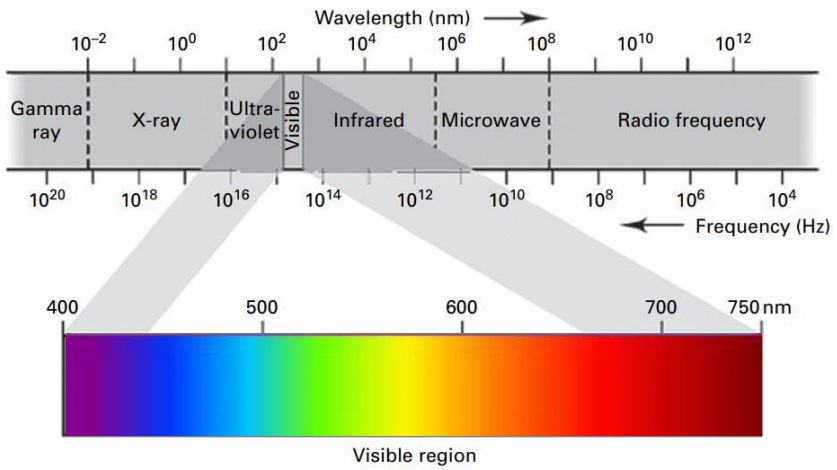
- Rule of thumb: The more complex the task, the more machine learning or artificial intelligence is involved
- In this course: Image processing foundations
  - How to represent images?
  - How to process images?
  - Which are the problems to solve?
- Not in this course: Methods of machine learning / AI

Context: Imaging systems by type of radiation (or field) used

- Electromagnetic waves carry the information captured in images.

- General rules:
  - The shorter the wavelength of a wave, the better it can reveal small details
  - The short the wavelength, the more energy carries a wave

- Types of radiation or fields used in imaging:
  - Electromagnetic waves: Radio, microwaves, infrared, visible light, ultraviolet, x-rays (soft or hard), γ-rays
  - Particle radiation: Neutrons, protons, electrons, heavy ions
  - Mechanical (acoustic) waves: Ultrasound (medical imaging), sonar (underwater imaging), seismic waves (geophysical imaging)



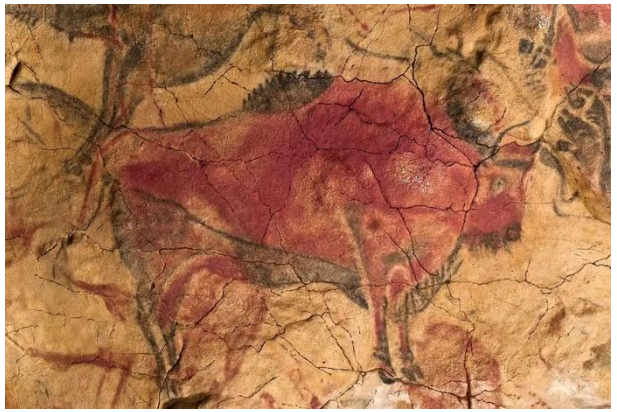
The electromagnetic spectrum arranged according to the energy of the photons, or the frequency of the waves.

A short history of images

Or rather, a ridiculously short history

A ridiculously short history of images

- Humans generate, record and transmit images to store and pass on information since the early days

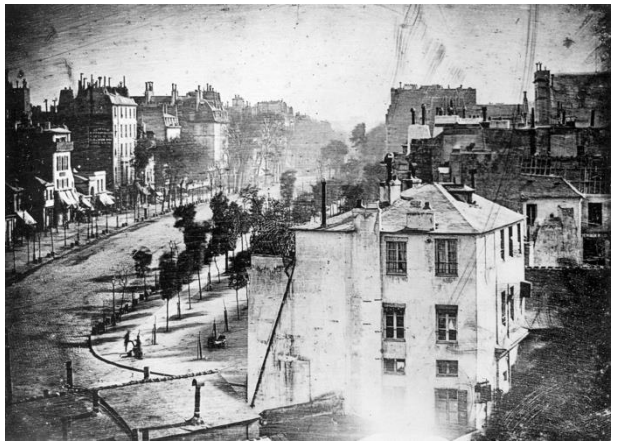


Painting of a bison from the Altamira cave, dating from around 36,000 years ago. (Source: Wikimedia)

A ridiculously short history of images

- Humans generate, record and transmit images to store and pass on information since the early days
- First photograph was pioneered in the 1820s by the Frenchmen Joseph Nicéphore Niépce and Louis Daguerre

1826: View from the Window at Le Gras by Niépce. Possibly the earliest surviving photograph. Source



1838: Boulevard du Temple, Paris. Photography by Louis Daguerre. Source

A ridiculously short history of images

- Humans generate, record and transmit images to store and pass on information since the early days
- First photograph was pioneered in the 1820s by the Frenchmen Joseph Nicéphore Niépce and Louis Daguerre
- First digital image was taken by Russel Kirsch in 1957 using a scanner
- First digital camera using a CCD (charge-coupled device) sensor from Willard S. Boyle and George E. Smith in 1969

1957: First digital image ever created, by Russel Kirsch. Source

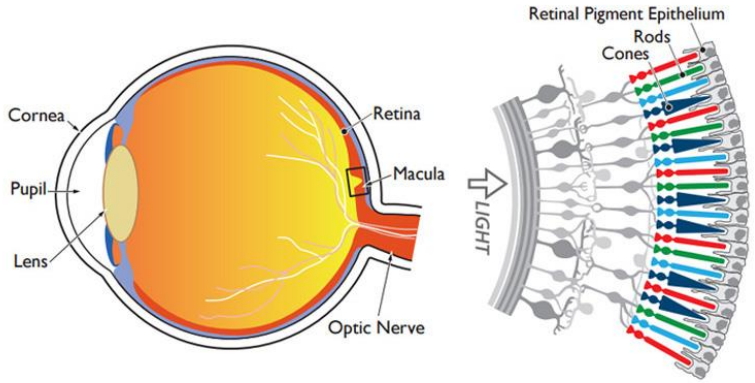
Human visual system

Take inspiration from nature

Human visual perception

- Up to about 60% of the brain activity are involved in processing visual stimuli
- More surprising facts about the human visual system
  - The image on the retina is upside-down
  - Retina is front-back inverted
  - Peripheral vision is low-res and nearly monochrome
  - The resolution of the eye is only 1 megapixel
  - The eye involves about 100 million photoreceptors
  - The eye has a special night-vision mode

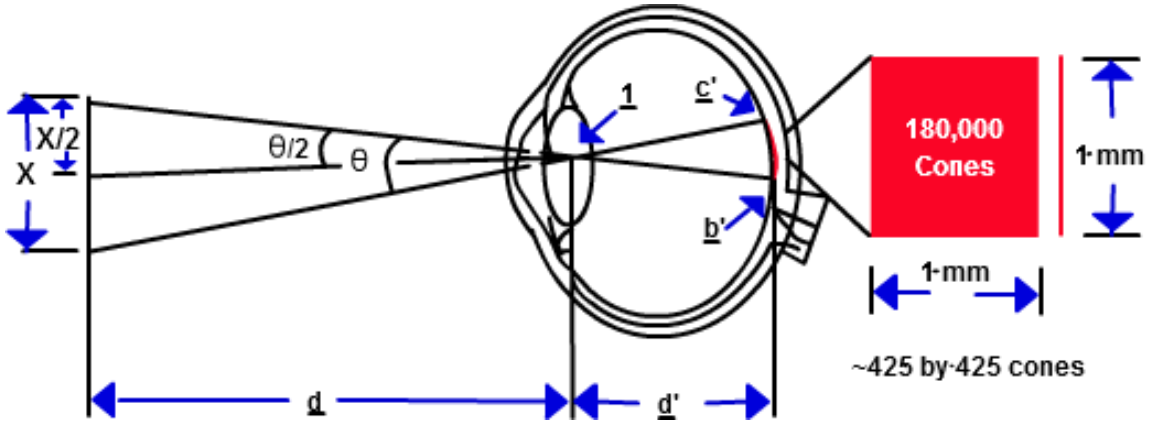
Close-up of the Retina



Illustrations: Link, Link

Resolution (human eye)

- The resolution of the human eye varies depending on several factors such as
  - the lighting conditions,
  - the contrast of the objects being observed
  - and the individual's visual acuity (Sehschärfe)
- Under optimal conditions, the human eye is estimated to have a resolution θ of around 1 arc minute
  - …which is equivalent to $\theta = 1/60° = 1/60 \cdot \pi/180$ rad
  - …which is equivalent to about x = 0.3mm for d = 1m
- See here for a reasoning.

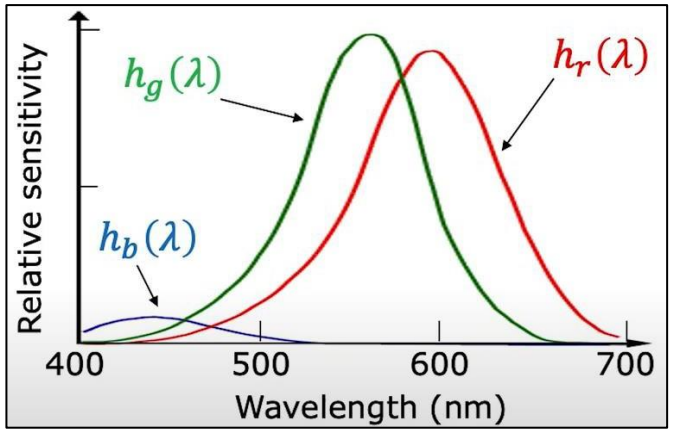


Human visual perception

- Rods and cones are specialized photoreceptor cells located in the retina of the eye.
  - Rods: very sensitive to light in general, particularly in low-light conditions, provide peripheral vision
  - Cones: Less sensitive to light, but provide the ability to perceive colors, enable high visual acuity (Sehschärfe)

- Three types of cones with different spectral sensitivity:
  - S-cones / blue
  - M-cones / green
  - L-cones / red

S: Short-wavelength
M: Medium-wavelength
L: Long-wavelength



«Quantum efficiency» of red, green and blue cones.

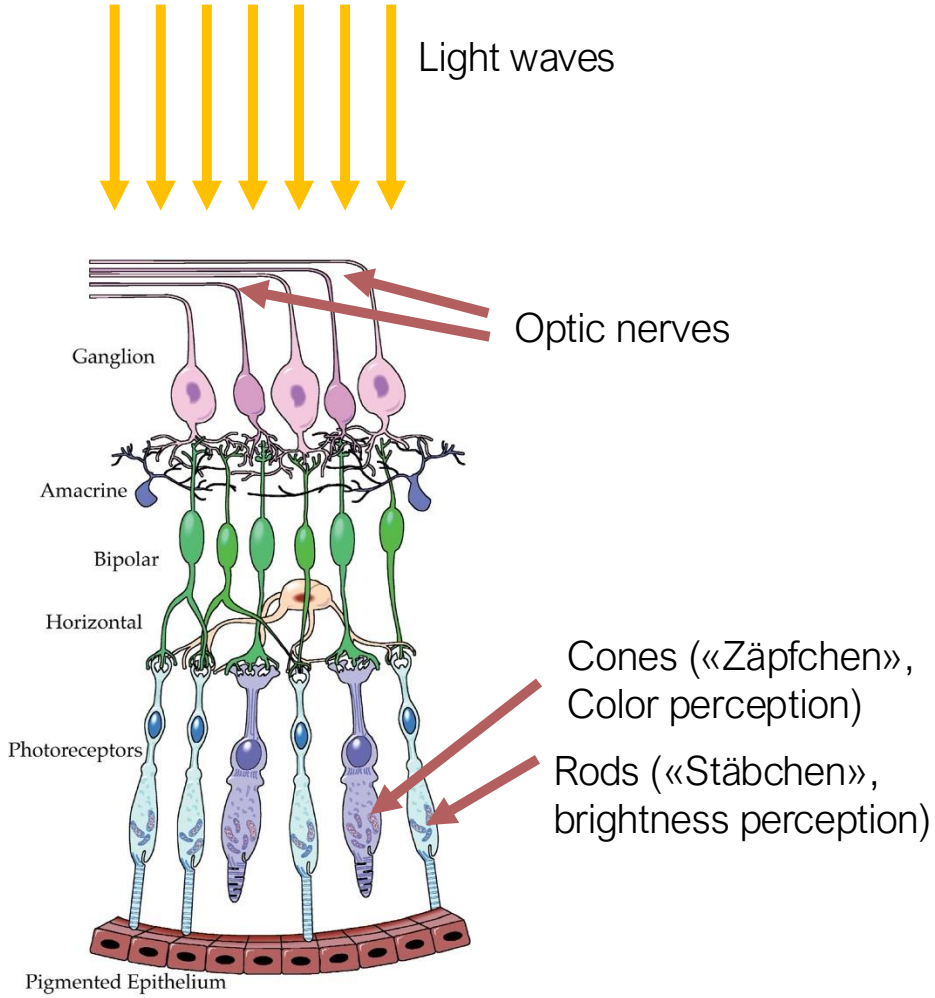


Illustration: Adapted from “Color Appearance”, Fairchild, 2005

Luminance vs. Chrominance

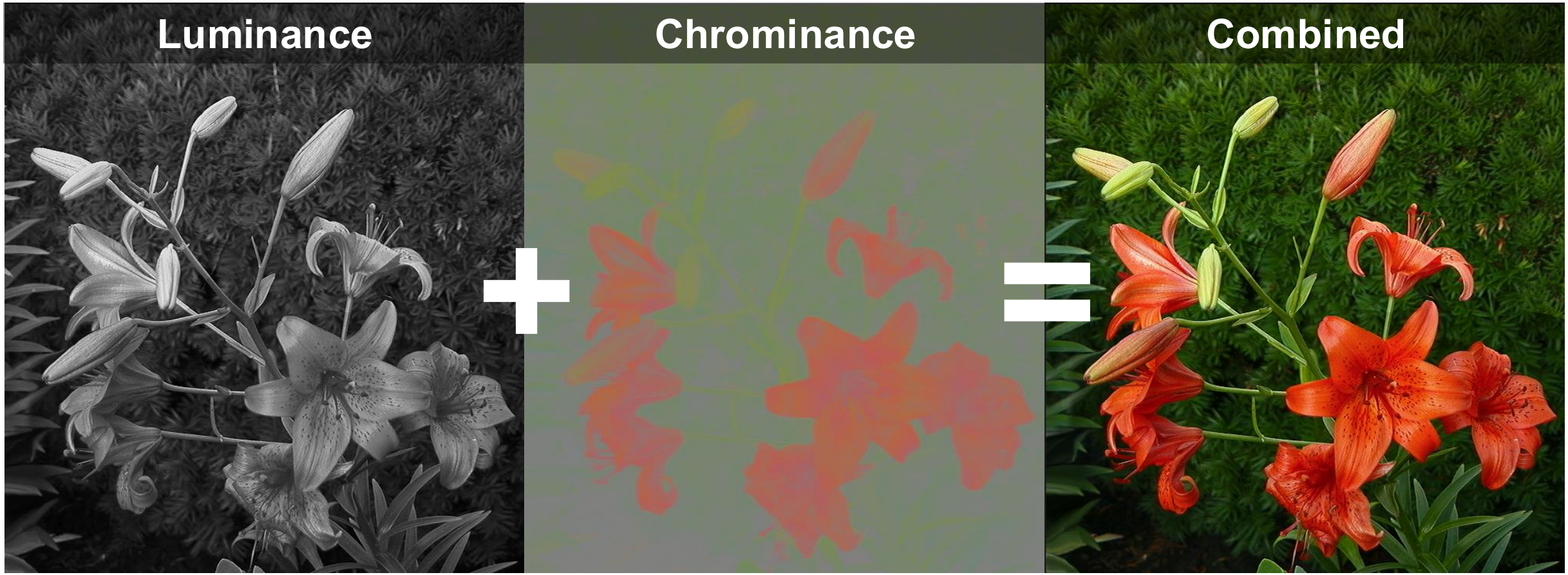


Image Source: Wikimedia

Luminance vs. Chrominance

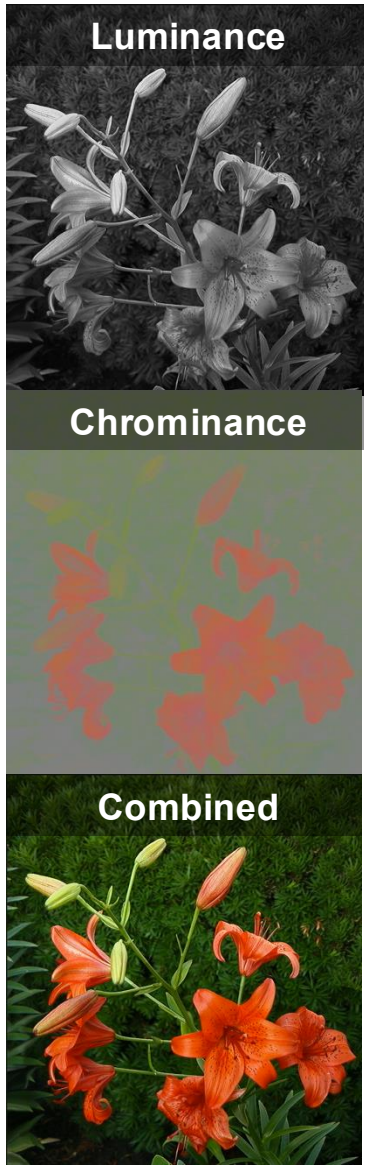
The human visual system processes **brightness** and **color** differently.

- **Luminance**: Brightness perception
  - Perceived light intensity of a scene
  - Mainly processed by rods
  - High sensitivity to contrast, edges, fine spatial details

  ⇒ Dominates our perception of **structure** and **sharpness**

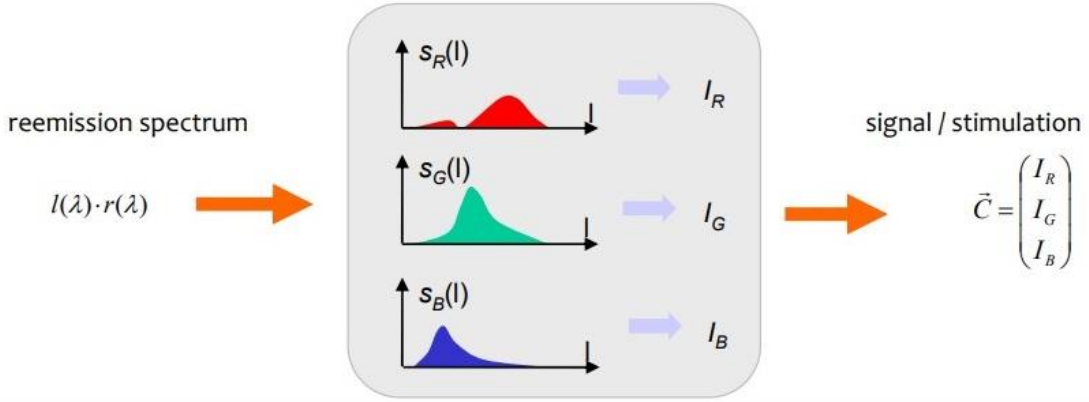
- **Chrominance**: Color perception
  - Represents color information (hue and saturation)
  - Processed by cones
  - Lower spatial resolution, less sensitive to fine detail

  ⇒ Adds **color appearance**, but contributes less to sharpness



Color perception

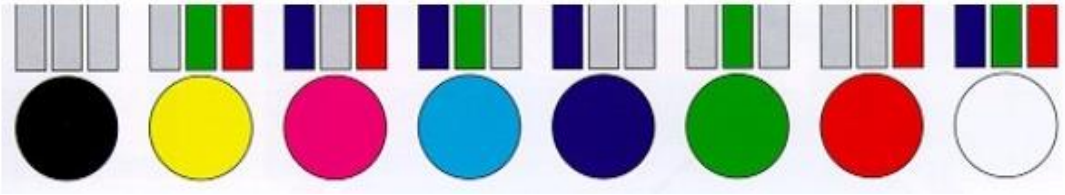
- Light propagation:
  - Light source spectrum: $l(\lambda)$
  - Reflectance / reemission spectrum: $r(\lambda)$
  - Sensitivity of cone receptors: $s_*(\lambda)$



- Result:
  - Signal / stimulation consists of three channels
  - Different stimulation patterns lead to the perception of different colors:

Stimuli

Perception



More on the human visual system

- Short intro to eye structure and function: Link
- Evolutionary perspective on vision, explaining some of its features: Link
- Color vision and additive color mixing: Link
- Human Vision vs. cameras – explores advanced features of our visual system: Link
- Old school explanatory video, addressing why the image is upside down and how eyesight defects occur. Link